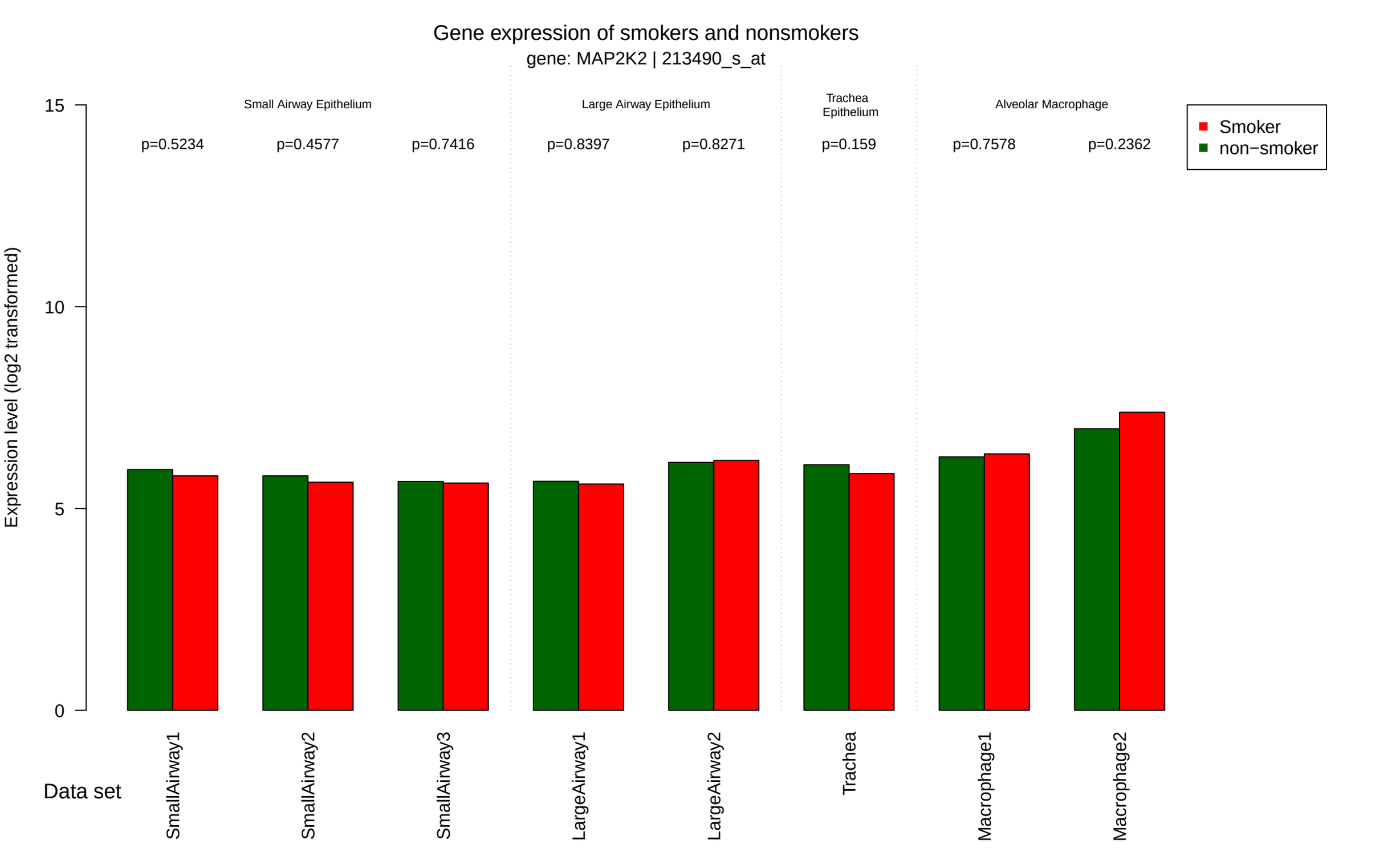
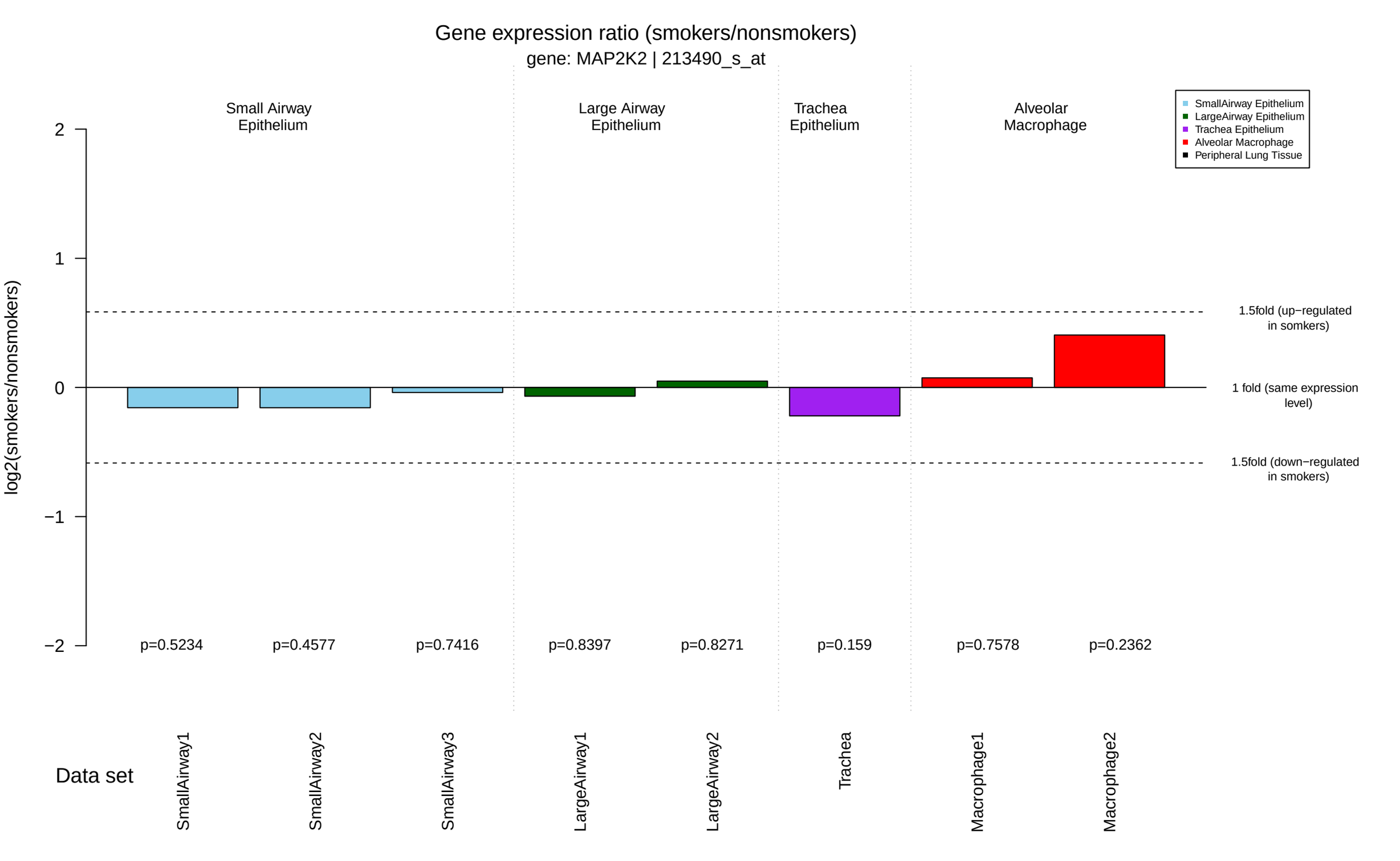
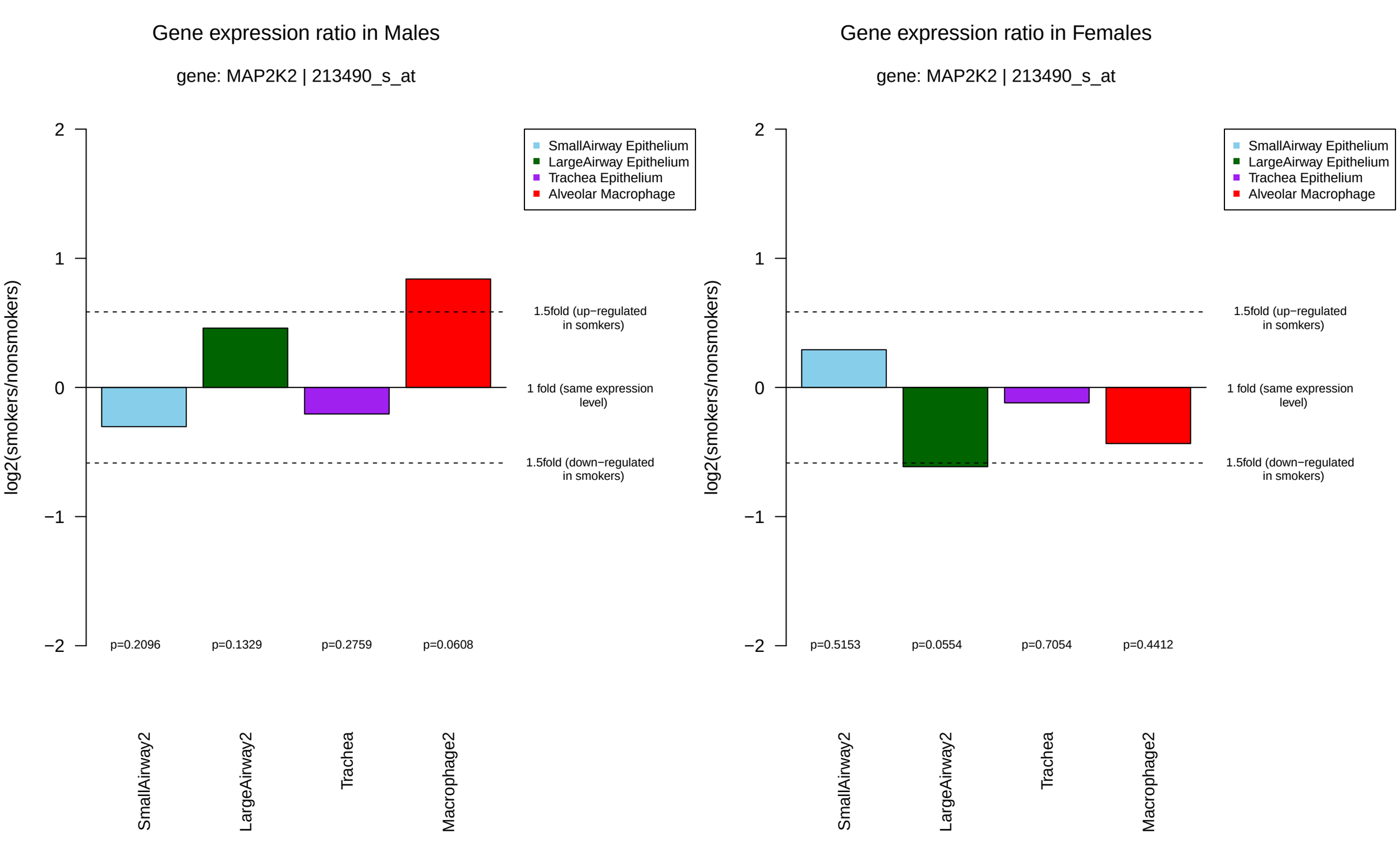
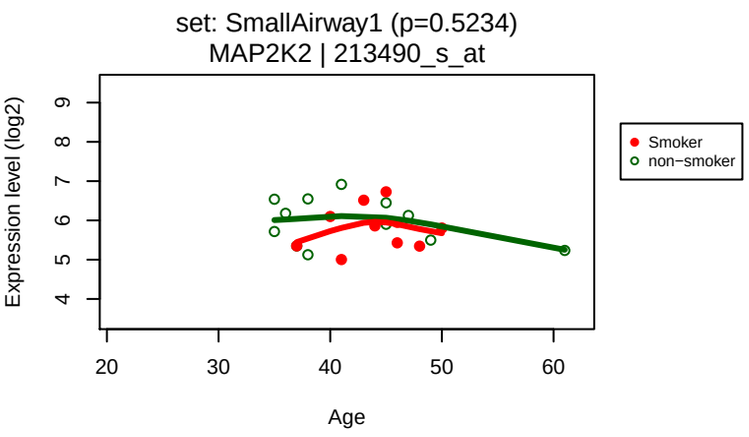
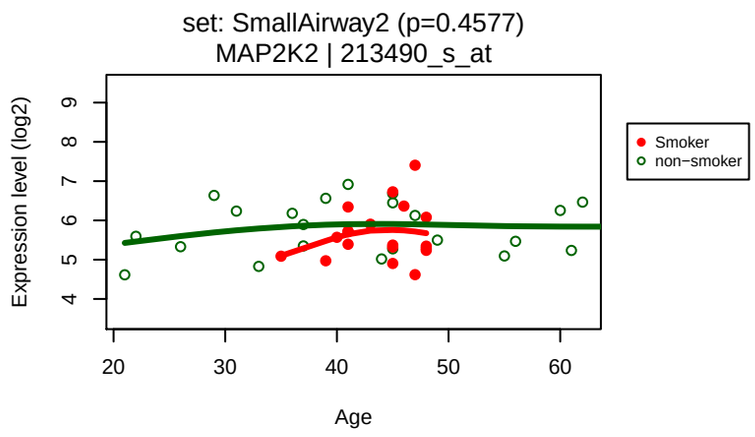
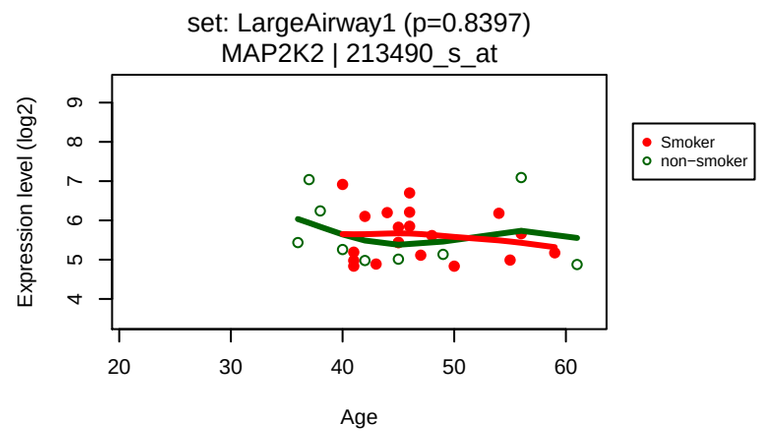

Gene expression of smokers and nonsmokers
gene: MAP2K2 | 213490_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5234
p=0.4577
p=0.7416
p=0.8397
p=0.8271
p=0.159
p=0.7578
p=0.2362
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

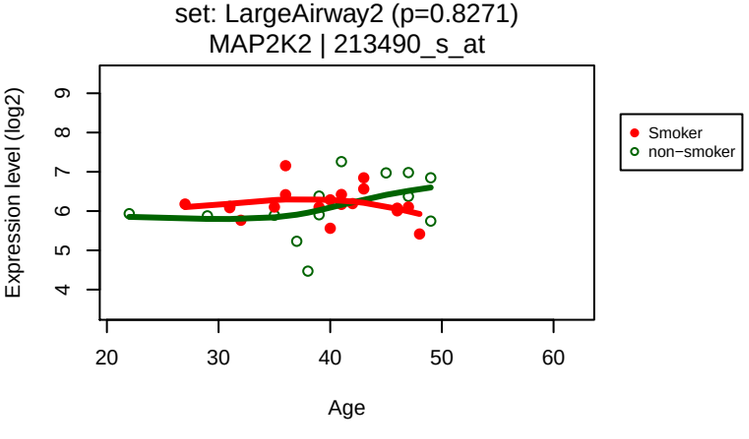

Gene expression ratio (smokers/nonsmokers)
gene: MAP2K2 | 213490_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5234
p=0.4577
p=0.7416
p=0.8397
p=0.8271
p=0.159
p=0.7578
p=0.2362
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

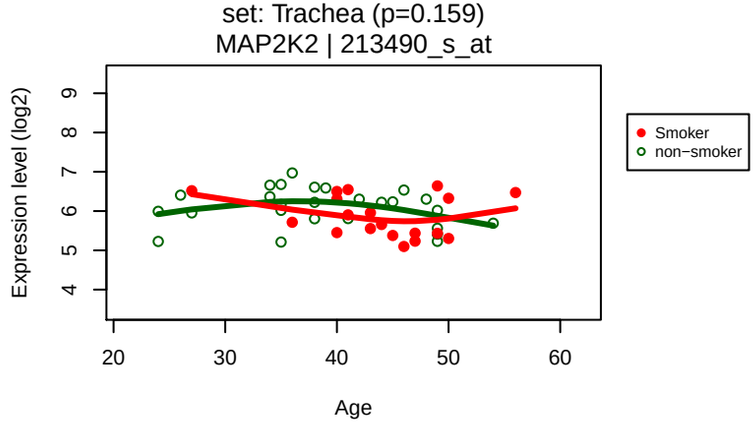

Gene expression ratio in Males
gene: MAP2K2 | 213490_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2096
p=0.1329
p=0.2759
p=0.0608
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAP2K2 | 213490_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5153
p=0.0554
p=0.7054
p=0.4412
SmallAirway2
LargeAirway2
Trachea
Macrophage2

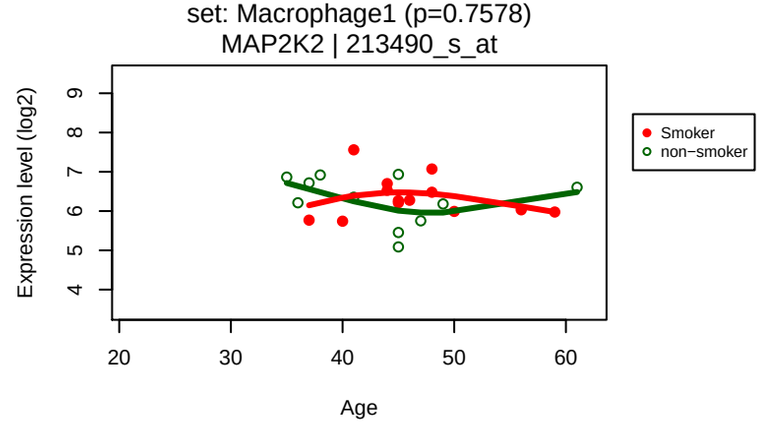

set: SmallAirway1 (p=0.5234)
MAP2K2 | 213490_s_at
Expression level (log2)
Age
Smoker
non−smoker

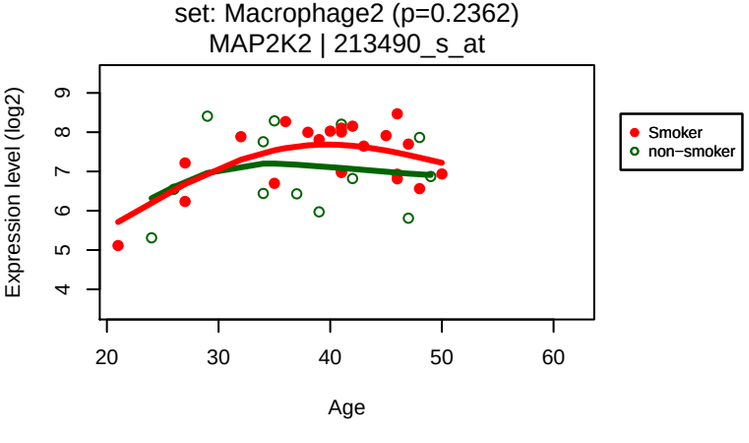

set: SmallAirway2 (p=0.4577)
MAP2K2 | 213490_s_at
Expression level (log2)
Age
Smoker
non−smoker

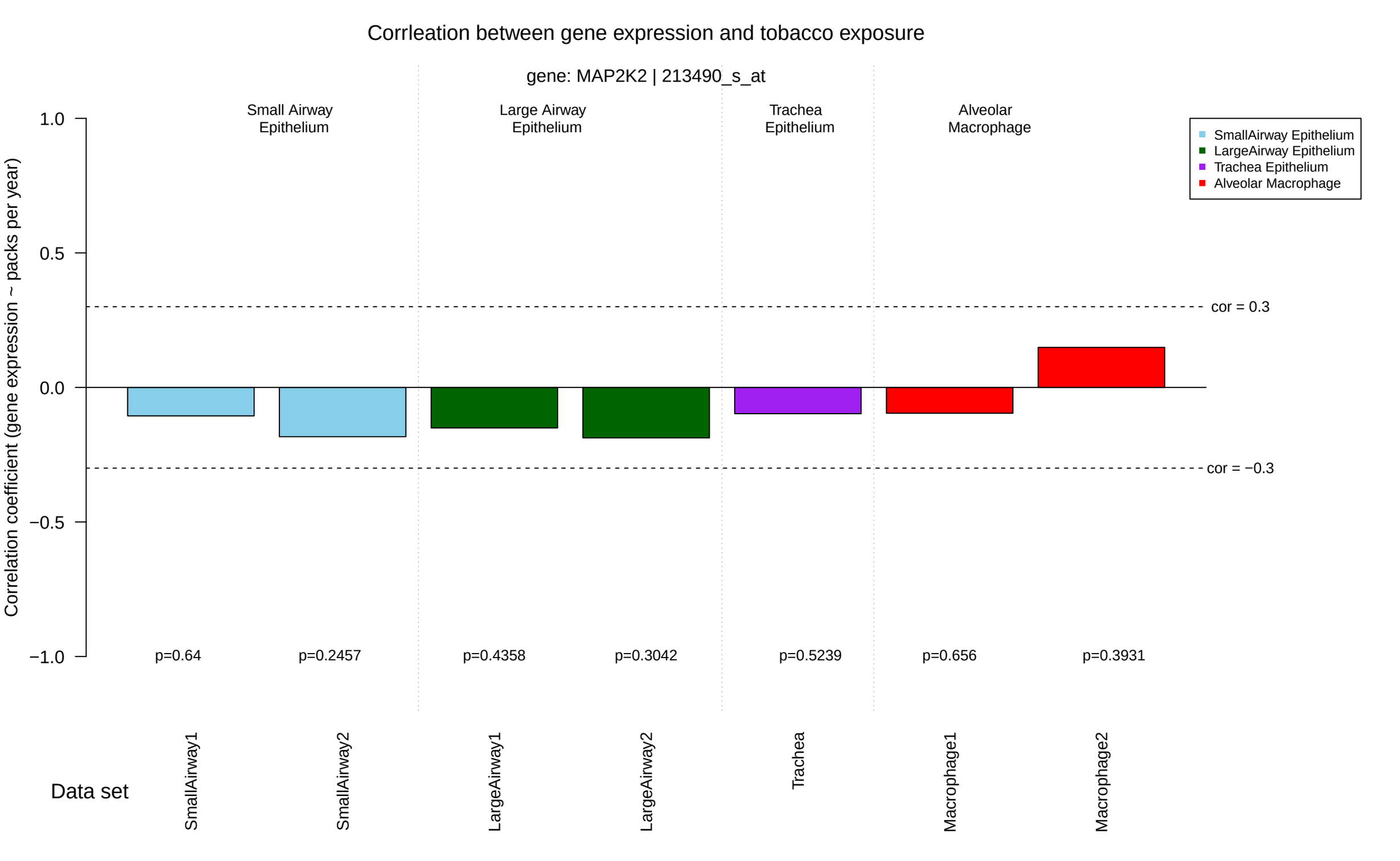

set: LargeAirway1 (p=0.8397)
MAP2K2 | 213490_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8271)
MAP2K2 | 213490_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.159)
MAP2K2 | 213490_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7578)
MAP2K2 | 213490_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2362)
MAP2K2 | 213490_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAP2K2 | 213490_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.64
p=0.2457
p=0.4358
p=0.3042
p=0.5239
p=0.656
p=0.3931
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2